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COMPENSATION OF SPACE-CHARGE EFFECTS AT TRANSITION

BY AN ASYMMETRIC Q-JUMP: A THEORETICAL STUDY

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SUMMARY

A Q_H -jump, sweeping $\gamma_{tr} \approx Q_H$ rapidly across the γ -distribution of the beam, is an effective method of reducing intensity dependent bunch blow up at transition energy.

The asymmetry introduced by longitudinal space-charge can be weakened by the asymmetry (mistiming) of the jump. The influence of a γ_{tr} -spread (resulting e.g. from transverse space-charge forces) decreases because transition is crossed rapidly. The peak momentum spread and thus aperture requirements are reduced because bunch shape oscillations are kept small.

Relations relevant to the design of a Q-jump system are derived.

1. INTRODUCTION

The Q-jump method was first proposed about 15 years ago and has frequently been reconsidered since, as a simple means of passing transition energy rapidly [1] and of alleviating longitudinal space-charge effects [2] at transition.

New interest in this old method was stimulated, when H. H. Umstätter [5] suggested, that transverse space-charge forces might introduce a spread in transition energy which leads to bunch blow up. It was then realised that Q-jumping is one of the few methods which avoid bunch blow up at transition in spite of the presence of a γ_{tr} -spread [3].

The design of a Q-jump system was therefore studied and experiments to test the method are at present being performed on the PS [4]. Parallel to the design work and in preparation for the experiments, some questions related to Q-jumping had to be studied theoretically.

These studies indicated that a delayed Q-jump (Fig. 1c) is much more effective in compensating bunch blow up at transition than earlier anticipated for a symmetric jump (Fig. 1b).

In the present report some results of both analytical and computer studies will be discussed.

2. PHYSICAL CONSIDERATIONS

(Why does space-charge lead to bunch blow up and why does a Q-jump provide compensation?)

2.1. Linear longitudinal space-charge forces

The influence of longitudinal space-charge forces is discussed in reports by Hereward and Sørensen (for a summary see e.g. [2]).

Below transition, space-charge increases the length of a matched bunch (Fig. 1a). Above transition space-charge decreases the equilibrium bunch length. A discontinuity occurs at transition. A bunch well matched below will be mismatched above transition, and its length starts to oscillate.

These oscillations "smear out" due to non-linearity and the phase space density decreases.

Now assume that by Q-jumping transition is rapidly swept across the energy distribution of the beam (Fig. 1b). This is equivalent to "cutting out" the transition region where bunches are short and space-charge forces thus strong. The mismatch is thus reduced as transition is passed with longer bunches.

A more effective compensation is possible by an "asymmetric Q-jump" as sketched in Fig. 1c. There the Q-jump is delayed so that the transition point is closer to the start of the jump. The equilibrium bunch length is then discontinuous at low intensities but continuous for some given intensity level. It is noted that this

second mechanism can be used to eliminate the discontinuity at transition whereas a symmetric jump only weakens it.

2.2. Transverse space-charge forces

The dependence of the Laslett Q-shifts on longitudinal position in a bunch may introduce a γ_{tr} -spread [5]. A particle riding at the front or the tail of a bunch will only "see" a small Q-shift whereas particles in the center experience the full Q-depression. A Q-spread, and, since $\gamma_{tr} \approx Q_H$, a γ_{tr} -spread of the order of the Laslett Q-shift may therefore be present. This is now referred to as the "Umstätter effect". Any γ_{tr} -spread leads to a spread

$$\Delta t = \frac{\Delta \gamma_{tr}}{\dot{\gamma}}$$

in transition time. Therefore, the timing of the rf-phase switch is incorrect for some groups of particles. The phase trajectories of these particles deform and apparent blow up occurs.

A Q-jump increases the crossing speed $\dot{\gamma} + |\dot{\gamma}_{tr}|$ and thus reduces the time-spread ^{*})

$$\Delta t = \frac{\Delta \gamma_{tr}}{\dot{\gamma} + |\dot{\gamma}_{tr}|}$$

3. ANALYTICAL CALCULATIONS

3.1. The Model

In the following sections we derive some results using the simplified $\eta = \frac{1}{\gamma_{tr}^2} - \frac{1}{\gamma^2}$ - law sketched in Fig. 3. In this model one has two regions with a different slope $\dot{\eta}$. Inside these regions the slope is constant and the "constant $\dot{\eta}$ - solution" [6] of the equations of motion can be applied. At the points where $\dot{\eta}$ is discontinuous, the solutions have to be matched. We only attempt to

^{*}) The timespread has to be compared to the characteristic time $T \propto (\dot{\gamma} + |\dot{\gamma}_{tr}|)^{-1/3}$. The bunch blow up is reduced by $\left(\frac{\dot{\gamma}}{\dot{\gamma} + |\dot{\gamma}_{tr}|} \right)^{2/3}$, see Sect. 3.

derive approximate relations leaving a quantitative analysis to be done by a computer programme which is discussed in Section 4.

3.2. The equations of motion

Synchrotron motion has been treated in many papers. We collect some results here which are needed in the following sections. Let us use $\Delta\beta\gamma$ rather than $\Delta E/w$ as momentum variable. Neglecting space-charge, phase motion is then determined by

$$\begin{aligned} \frac{d}{dt} (\Delta\beta\gamma) &= \frac{\dot{\beta\gamma}}{\sin\varphi_s} (\sin\varphi - \sin\varphi_s) \\ \frac{d\varphi}{dt} &= \frac{\eta \omega_{rf}}{\gamma} \Delta\beta\gamma \end{aligned} \quad (3.1)$$

Following Hereward and Sørensen we normalize all times to the machine constant T (see Appendix 2). Further we linearize writing $\sin\varphi - \sin\varphi_s \approx \cos\varphi_s (\varphi - \varphi_s) = \cos\varphi_s \lambda$ where λ is the deviation from the synchronous phase. We neglect the time variation of all parameters except the change of η . Combining (3.1) into a second order equation for λ one has [6]

$$\begin{aligned} \frac{d}{dx} \left(\frac{\eta'}{\eta} \frac{d\lambda}{dx} \right) + \text{sign}(\eta - \eta_{SW}) \lambda &= 0 \\ \frac{d\lambda}{dx} &= \frac{\eta}{\eta'} \frac{\Delta\beta\gamma}{k(3/2)^{2/3}} \end{aligned} \quad (3.2)$$

Here $x = t/T$ is the normalised time, $\eta' = \frac{d\eta}{dx} = \dot{\eta} T, k = (2/3)^{2/3} \frac{\gamma}{\eta \omega_{rf} T^2}$, η_{SW} is the $\eta = \frac{1}{\gamma_{tr}^2} - \frac{1}{\gamma^2}$ - value at the point, where the rf-phase is switched, the time T (Appendix 2) is a measure of the half width of the non-adiabatic region.

Linear longitudinal space-charge forces can be included by modifying the rf-focusing term $\beta\dot{\gamma} |\cos\varphi_s| \rightarrow \beta\dot{\gamma} (1 + \text{sign}(\eta - \eta_{SW}) \eta_{SC})$

and thus $\text{sign}(\eta - \eta_{\text{SW}}) \rightarrow \text{sig}(\eta - \eta_{\text{SW}})(1 + \text{sign}(\eta - \eta_{\text{SW}})\eta_{\text{SC}})$ in (3.2). The space-charge parameter $\eta_{\text{SC}}(x)$ is calculated using the modified bunch length and is thus different from η_0 [6].

We now restrict ourselves to a linear change of $\frac{1}{v^2} - \frac{1}{v_{\text{tr}}^2}$. Neglecting space-charge, the solution of (3.2) may be written in terms of Bessel functions of argument $y = 2/3(x)^{3/2}$. We will use the phase amplitude form of the Bessel functions [7]

$$J_V = M_V(x) \cos \theta_V(x)$$

$$N_V = M_V(x) \sin \theta_V(x)$$

with

$$\theta_V = \text{atan} \left(\frac{N_V(x)}{J_V(x)} \right)$$

$$M_V^2 = J_V^2 + N_V^2(x)$$

The constant $\dot{\eta}$ solution can then be written as

$$\begin{aligned} \lambda &= B y^{2/3} M_{2/3}(y) \cos \psi \\ \Delta \beta \gamma &= k B y^{1/3} M_{-1/3}(y) \sin(\psi - \alpha) \end{aligned} \quad (3.3)$$

Here $\psi = \theta_{2/3}(y) + \delta$, $\alpha = \frac{\pi}{2} + \theta_{-1/3}(y) - \theta_{2/3}(y)$, B and δ are constants given by the initial conditions. Note that the phase space area enclosed by a trajectory $B = \text{const}$ is $A = 2 k B^2$.

In Fig. 4, the equilibrium bunch length and height

$$\begin{aligned} \hat{\lambda} &= B y^{2/3} M_{2/3}(y) \\ \Delta \hat{\beta} \gamma &= B k y^{1/3} M_{-1/3}(y) \end{aligned}$$

is reproduced as a function of x . In particular, at transition one has

$$\mathcal{A}(0) \approx B \cdot 0.68 \cos\psi \quad (3.4)$$

$$\Delta\beta\gamma(0) \approx kB \cdot 1.07 \sin(\psi - 150^\circ)$$

And for $y \gg 1$

$$\mathcal{A} = B \left(\frac{2}{\pi}\right)^{1/2} y^{1/6} \cos(y + \delta_1) \quad (3.5)$$

$$\Delta\beta\gamma = kB \left(\frac{2}{\pi}\right)^{1/2} y^{-1/6} \sin(y + \delta_1)$$

In the presence of space-charge, the constant $\dot{\eta}$ -solution is no longer valid. In the adiabatic region the scaling of bunch length and height can be obtained from (3.2) by the WKB -method:

$$\begin{aligned} \hat{\mathcal{A}}_{SC} &= \hat{\mathcal{A}} (1 + \text{sign}(x - x_{SW}) \eta_{SC})^{-1/4} \\ \Delta\hat{\beta}\gamma_{SC} &= \Delta\hat{\beta}\gamma (1 + \text{sign}(x - x_{SW}) \eta_{SC})^{1/4} \end{aligned} \quad (3.6)$$

Here we denote by x_{SW} the time at which the rf-phase is switched.

Near transition, the bunch length changes slowly. As an approximation $\eta_{SC}(x) \approx \eta_{SC}(0)$ can then be assumed to be constant and one has

$$\begin{aligned} \hat{\mathcal{A}}_{SC} &= \hat{\mathcal{A}} \left[1 + \text{sign}(x - x_{SW}) \eta_{SC} \right]^{-1/3} \\ \Delta\hat{\beta}\gamma_{SC} &= \Delta\hat{\beta}\gamma \left[1 + \text{sign}(x - x_{SW}) \eta_{SC} \right]^{1/3} \end{aligned} \quad (3.7)$$

We can now establish the relation between $\eta_{SC}(x)$ and $\eta_0(x)$ for a matched bunch. Since $\eta_{SC} \propto \hat{\mathcal{A}}_{SC}^{-3}$, $\eta_0 \propto \hat{\mathcal{A}}^{-3}$ [6] we have near transition (using (3.7))

$$\eta_o = \frac{\eta_{SC}}{1 + \text{sign}(x - x_{SW}) \eta_{SC}} \quad (3.8)$$

and in the adiabatic region (using (3.6))

$$\eta_o = \frac{\eta_{SC}}{1 + \text{sign}(x - x_{SW}) \eta_{SC}^{3/4}} \quad (3.9)$$

Note that for $\eta_o \gg 1$ and $x < x_{SW}$

$$\eta_{SC} \approx 1 - \eta_o^{-1} \quad \text{near transition} \quad (3.10)$$

$$\eta_{SC} \approx 1 - \frac{3}{4} \eta_o^{-1} \quad \text{in the adiabatic region}$$

and for $\eta_o \ll 1$:

$$\eta_{SC} \approx \eta_o \quad (3.11)$$

3.3. Infinitely fast jump

(How far has one to jump)

For an infinitely fast jump, the η_{SC} -value at transition is determined by the η -value at the starting point of the jump.

The corresponding x -value is obtained, noting (Fig. 2) that

$$\delta\eta = \eta'(x_2 - x_1)$$

Therefore, the jump width is in x -units

$$x_2 - x_1 = \frac{\delta\eta}{\eta'} \approx \frac{\delta\gamma_{tr}}{\gamma'} \quad (3.12)$$

For a symmetric jump ($|x_1| = x_2$)

$$|x_1| = \frac{1}{2} \frac{\delta \gamma_{tr}}{\gamma'}$$

For a symmetric jump the improvement in η_0 can then be read from the $\eta_0(x)$ -curve Fig. 5.

Neglecting rotation of the phase ellipses near transition, $\eta_0(x_1)$ has now the same significance as $\eta_0(0)$ in the normal case (no Q-jump). One can then estimate the blow up in area using [6] :

$$\frac{A_2}{A_1} \approx 1 + 1.4 \eta_0(x_1) ; \eta_0(x_1) \lesssim 1$$

We can thus estimate the improvement obtainable with a symmetric jump. Let us now estimate the width of an asymmetric jump required for perfect compensation. We assume that the jump starts close to transition, so that $|x_1| \ll 1, x_2 \gg 1$.

The equilibrium bunch length at the edges of the jump region is then, by virtue of (3.8), (3.9) and (3.4), (3.5)

$$\hat{\lambda}_{SC}(x_1) \approx \lambda(0) (1 - \eta_{SC}(0))^{-1/3}$$

$$\hat{\lambda}_{SC}(x_2) \approx 1.07 \lambda(0) x_2^{1/4} (1 + \eta_{SC}(0))^{-1/4}$$

Requiring continuity $\hat{\lambda}_{SC}(x_1) = \hat{\lambda}_{SC}(x_2)$ the minimum jump width required is :

$$x_2 - x_1 \approx x_2 \approx 0.76 \frac{(1 + \eta_{SC}(0))}{(1 - \eta_{SC}(0))^{4/3}}$$

Or expressing $\eta_{SC}(0)$ by $\eta_0(0)$ by means of (3.10) or (3.11)

$$x_2 \approx 1.5 \eta_o^{4/3}(0) \quad (\eta_o(0) \gg 1)$$

$$x_2 \approx 0.76(1+2.3 \eta_o(0)) \quad (\eta_o(0) \ll 1)$$

Continuity of bunch length is not a sufficient condition for avoiding mismatch, since the phase ellipses are rotated near transition. However, this rotation can partly be compensated by "mistiming" of the rf-phase switch. It is thus concluded that a fast asymmetric jump with

or

$$(x_2 - x_1) = \frac{\delta \gamma_{tr}}{\gamma} \gtrsim 1.5 \eta_o^{4/3}(0) \quad (\eta_o(0) \gg 1)$$

(3.14)

$$\frac{\delta \gamma_{tr}}{\gamma} \gtrsim 0.76(1+2.3 \eta_o(0)) \quad (\eta_o \ll 1)$$

will provide almost perfect compensation of the linear longitudinal effect. Partial compensation can be obtained with a smaller jump.

As an example take $\frac{\delta \gamma_{tr}}{\gamma} \approx 5$ obtainable in the CPS or the 300 GeV-booster with a Q-jump of about 0.25. Then full compensation is possible for η_o -values $\eta_o(0) \lesssim 2.5$.

Finally we want to calculate the maximum momentum spread $\hat{\Delta\beta\gamma}$. Without Q-jump the equilibrium bunch height just after transition is (see 3.7)

$$\hat{\Delta\beta\gamma}_{SC} = \hat{\Delta\beta\gamma}_o (1+\eta_{SC})^{1/3}$$

Therefore, even if the bunch could be kept matched the momentum width would increase with intensity.

Without compensation the bunch height oscillates and the peak excursion occurring shortly after transition can be estimated [8] using

$$(\Delta\beta\gamma_{SC})_{\max} \approx (\Delta\beta\gamma_o)_{\max} 1.18 \eta_o(0) ; \quad \eta_o(0) \gtrsim 1$$

Now assume that by means of an asymmetric jump the discontinuity in bunch height is removed and bunch shape oscillations are avoided. If the jump starts at $x_1 \approx 0$, the bunch height in the jump region which determines the maximum value is (from 3.7) :

$$\begin{aligned} (\Delta\beta\gamma_{SC})_{\max} &= (\hat{\Delta\beta\gamma}_o(0)) (1 - \eta_{SC})^{1/3} \\ &\approx \hat{\Delta\beta\gamma}_o(0) \eta_o^{-1/3}(0) \quad \eta_o \gg 1 \end{aligned} \tag{3.15}$$

Thus the peak bunch height is even reduced beyond the low intensity value. It seems that Q-jumping is the only method offering this advantage which may be important for aperture considerations.

Note, however, that (3.15) is only valid for an infinitely fast jump; the more general relation

$$\begin{aligned} (\Delta\beta\gamma_{SC})_{\max} &\approx \hat{\Delta\beta\gamma}_o(0) \cdot \eta_o^{-1/3} \left[1 + 0.63 \eta_o(0) \cdot \frac{\delta \gamma_{tr}}{\gamma} \right. \\ &\quad \left. \left(\frac{\dot{\gamma}}{\dot{\gamma} - \dot{\gamma}_{tr}} \right)^{2/3} \right] \quad (\eta_o(0) \gg 1) \end{aligned}$$

will be derived in Section 4.4.

3.4. Required speed of the Q-jump

With a fast enough γ_{tr} decrease, the bunch shape is the same at the start and at the end of the jump. How fast has one to jump to be so that this feature is preserved?

Looking at the jump region one has to normalize time to

$$T_j \equiv T \left(\frac{\dot{\gamma}}{\dot{\gamma} - \dot{\gamma}_{tr}} \right)^{-1/3}$$

Note that on both time-scales the jump starts at a given distance $\Delta\eta = \dot{\eta}t = \dot{\eta}_j t$ from transition. Therefore on the T_j -scale, the start and the end of the jump region have the coordinates :

$$\begin{aligned} x_{1j} &= x_1 \left(\frac{\dot{\gamma}}{\dot{\gamma} - \dot{\gamma}_{tr}} \right)^{2/3} \\ x_{2j} &= x_2 \left(\frac{\dot{\gamma}}{\dot{\gamma} - \dot{\gamma}_{tr}} \right)^{2/3} \end{aligned} \tag{3.16}$$

Inside the jump region one is therefore "close to transition" ($x_j \ll 1$) and we need the solution of the equations of motion for small x . Let us write (3.2) in the form

$$\frac{d}{dx} \left(\frac{1}{x} \frac{d\lambda}{dx} \right) = -\lambda \operatorname{sign}(x - x_{SW}) (1 + \eta_{SC} \operatorname{sign}(x - x_{SW}))$$

$$\frac{d^2}{dx^2} (\Delta\beta\gamma) = -\Delta\beta\gamma x \operatorname{sig}(x - x_{SW}) (1 + \eta_{SC} \operatorname{sig}(x - x_{SW}))$$

Here space-charge is introduced assuming that $\eta_{SC} = \eta_{SC}(x_1)$ is constant in the jump region. A solution for small x is obtained by iteration, inserting approximate values for λ and $\Delta\beta\gamma$ on the right hand side :

$$\begin{aligned}\lambda &= \lambda(0) + c_1 x^2 + c_2 x^3 + \dots \\ \Delta\beta\gamma &= \Delta\beta\gamma(0) + d_1 x + d_2 x^3 + \dots\end{aligned}\tag{3.17}$$

The constants c_i and d_i can be determined using the second equation (3.2)

$$\begin{aligned}c_1 &= \frac{1}{2} \frac{\Delta\beta\gamma(0)}{(3/2)^{2/3} k} \\ d_1 &= - \left(\frac{3}{2}\right)^{2/3} k \lambda(0) \operatorname{sign}(x - x_{SW}) \left[1 + \eta_{SC} \operatorname{sign}(x - x_{SW}) \right] \\ c_2 &= \frac{1}{3} \frac{d_1}{k(3/2)^{2/3}} \\ d_2 &= \frac{1}{2} \frac{d_1}{k(3/2)^{2/3}} \frac{\Delta\beta\gamma(0)}{\lambda(0)}\end{aligned}\tag{3.18}$$

Retaining only terms up to first order in x :

$$\begin{aligned}\lambda &= \lambda(0) \\ \Delta\beta\gamma &= \Delta\beta\gamma_0 + d_1 x\end{aligned}\tag{3.19}$$

Using this approximate solution the optimum switching time x_{SWj} can be found. Going from $x = x_{1j}$ to $x = x_{2j}$, $\Delta\beta\gamma$ changes by

$$d(\Delta\beta\gamma) = \left(\frac{3}{2}\right)^{2/3} k \lambda_0 \left[(1 - \eta_{SC})(x_{SWj} - x_{1j}) - (1 + \eta_{SC})(x_{2j} - x_{SWj}) \right] \tag{3.20}$$

Note that this change is zero if :

$$\boxed{x_{SW} = \frac{x_2 + \frac{1-\eta_{SC}}{1+\eta_{SC}} x_1}{1 + \frac{1-\eta_{SC}}{1+\eta_{SC}}}} \quad (3.21)$$

or expressing $\eta_{SC}(x_1)$ by $\eta_o(x_1)$ using (3.10)

$$\boxed{x_{SW} = \frac{x_2 + \frac{1}{2\eta_o(x_1)} x_1}{1 + \frac{1}{2\eta_o(x_1)}}} \quad (3.22)$$

$$\begin{aligned} \eta_o(x_1) &>> 1 \\ x_1 &<< 1 \end{aligned}$$

Thus even with a finite speed of the Q-jump the bunch shape can be made equal at the start and the end of the jump by choosing the rf switching point somewhat "too late".

Note from (3.21) that

$$x_{SW} = \frac{x_2 + x_1}{2} \quad \text{for } \eta_o \rightarrow 0$$

$$x_{SW} = x_2 \quad \text{for } \eta_o >> 1$$

We can now find the speed required by looking at the higher order terms in (3.18).

Retaining the quadratic term in (3.17) the blow up in area is

$$\frac{A_2}{A_1} \approx \frac{\hat{\delta}^2(x_{2j})}{\hat{\delta}^2(x_{1j})} \approx 1 + \frac{2c_1}{\hat{\delta}(0)} (x_{2j}^2 - x_{1j}^2)$$

The index j is used here to indicate that all times are normalized to the T-scale of the jump region.

Noting from (3.4) that for $x \ll 1$:

$$\hat{\lambda}_{(0)} \approx 0.684 (1-\eta_{SC})^{-1/3}$$

$$\Delta\beta\gamma_{(\lambda=\hat{\lambda})} \approx 1.07 (1-\eta_{SC})^{1/3} k \sin 30^\circ$$

one has from (3.18) :

$$\frac{c_1}{\hat{\lambda}_{(0)}} \approx 0.3 (1-\eta_{SC})^{2/3}$$

The factors $(1-\eta_{SC})^{1/3}$ take into account bunch lengthening (3.7) due to space-charge. Returning to the time scale outside the jump region (using 3.16) and solving for $\dot{\gamma}_{tr}$ we arrive at the criterion for the jump speed :

$$\begin{aligned} \frac{|\dot{\gamma}_{tr}|}{\dot{\gamma}} &\gtrsim 0.68 (1-\eta_{SC})^{1/2} \left(\frac{x_2^2 - x_1^2}{(A_2/A_1)-1} \right)^{3/4} - 1 \\ &\approx 0.68 \eta_0^{-1/2}(x_1) \left(\frac{x_2^2 - x_1^2}{A_2/A_1-1} \right)^{3/4} - 1 ; \quad \eta_0(x_1) \gg 1 \\ &\quad \quad \quad A_2/A_1 - 1 \ll 1 \end{aligned} \tag{3.23}$$

A similar criterion can be obtained assuming a given (total) spread $\Delta\gamma_{tr}$ in transition energy. Taking into account the variation of the linear term in (3.19) one obtains :

$$\boxed{\frac{A_2}{A_1} \approx 1 + 1.7 (1-\eta_{SC})^{-2/3} \left(\frac{\dot{\gamma}}{\dot{\gamma} + |\dot{\gamma}_{tr}|} \right)^{2/3} \frac{\Delta\gamma_{tr}}{2\gamma'}} \quad (3.24)$$

or solving for $\frac{\dot{\gamma}_{tr}}{\dot{\gamma}}$:

$$\boxed{\begin{aligned} \frac{|\dot{\gamma}_{tr}|}{\dot{\gamma}} &\gtrsim 2.2 (1-\eta_{SC})^{-1} \left(\frac{\Delta\gamma_{tr}/(2\gamma^1)}{(A_2/A_1)-1} \right)^{3/2} - 1 \\ &\approx 2.2 \eta_0(x_1) \left(\frac{\Delta\gamma_{tr}/(2\gamma^1)}{(A_2/A_1)-1} \right)^{3/2} - 1 ; \quad \eta_0(x_1) \gg 1 \\ &\quad A_2/A_1 - 1 \ll 1 \end{aligned}} \quad (3.25)$$

Note that (3.24) is also applicable to the situation without Q-jump ($\dot{\gamma}_{tr}=0$). However, estimating the influence of a γ_{tr} -spread one has to remember, that A_2/A_1 obtained from (3.24) is the blow up in phase space area of some groups of particles and thus only a somewhat unprecise measure of the bunch blow up.

As an example we take

$$A_2/A_1 = 1.15 \quad , \quad \eta_0(x_1) = 1.5 \quad ,$$

$$\frac{\Delta\gamma_{tr}}{2\gamma} = 0.3 \quad , \quad x_1 = 0 \quad , \quad x_2 = 4$$

and obtain

$$\frac{|\dot{\gamma}_{tr}|}{\dot{\gamma}} \gtrsim 8.5 \quad \text{from (3.25)}$$

$$\frac{|\dot{\gamma}_{tr}|}{\dot{\gamma}} \gtrsim 16.5 \quad \text{from (3.23)}$$

Finally the peak momentum deviation may be estimated looking again at the dx -term in (3.19) and inserting x_{SW} as obtained from (3.21) or (3.22) into (3.20). A correction factor to the momentum deviation (3.15) is thus obtained which may be written as

$$\frac{(\Delta\beta\gamma)_{\max}}{(\Delta\beta\gamma_{(\dot{\gamma}_{tr} \rightarrow \infty)})_{\max}} = 1 + 0.63 \eta_0(x_1) \frac{\delta\gamma_{tr}}{\gamma^1} \left(\frac{\dot{\gamma}}{\dot{\gamma} - \dot{\gamma}_{tr}} \right)^{2/3}$$

$$\eta_0(x_1) \gg 1$$

The requirement to keep the momentum spread small may introduce a further restriction on the speed of the Q-jump.

We have assumed that it is desirable to have the same bunch shape at the start and the end of the jump. However, if $|x_1| < 1$, the phase space is rotated at the start of the jump and this rotation has to be compensated if the bunch is to be matched at $x_2 > 1$. Some compensation is possible by choosing the timing of the rf-phase-switch to be at a somewhat different value from that derived above (3.21). All above relations are approximate since they neglect this "mistiming" of the phase switch.

Further the assumption $\eta_{SC}(x) = \text{const.}$ is only consistent in the linear approximation (3.19). Thus (3.23), which was obtained retaining the quadratic term in (3.17) is only exact for $\eta_0 \rightarrow 0$.

4. NUMERICAL RESULTS

A computer programme written by A. Sørenssen [6] was modified to be applicable for any $\eta_{(x)} = \frac{1}{\gamma_{tr}^2} - \frac{1}{\gamma^2}$ low. It integrates the equations of motion including linear longitudinal space-charge forces.

The following "envelope equations" for the bunch length $\theta \propto \hat{\delta}$ is used (derivation see Appendix 2) in the modified version :

$$\frac{\eta'}{\eta} \frac{d\theta}{dx} = P$$

$$\frac{dP}{dx} = - \text{sig} (\eta - \eta_{SW}) \theta - \frac{n}{\theta^2} + \frac{\eta}{\eta' \theta^3}$$

The notation is the same as that used in ref. [6] with the exception that we use n instead of N for the space-charge parameter. All x -values are normalized to the time constant T as calculated with $\dot{\gamma}_{tr} = 0$ (no Q-jump) ; $\eta' = \frac{d\eta}{dx} \approx \frac{2}{\gamma_{tr}^3} \dot{\gamma} T$; n is related to $\eta_0(0)$ by $n = 0.772 \eta_0(0)$; θ is proportional to the bunch length and P is related to the bunch height by

$$(\Delta\beta\gamma)^2 \propto (P^2 - \frac{1}{\theta^2})$$

Note that for $\eta = \eta'x$ the above equations are those used in ref. [6].

The programme is used together with a minimisation routine which finds the optimum timing of the Q-jump and the rf-phase switch.

Some results for a machine with $\gamma' = 0.05$, $\delta\gamma_{tr} = 0.25$ are presented in table 1 (note that mainly the parameter $\delta\gamma_{tr}/\gamma'$ is of importance for comparison between different machines). In Fig. 7 we reproduce the increase in bunch area as function of the speed of the jump.

5. CONCLUSION

A "delayed Q-jump" (Fig. 1c) is a very effective means of reducing space-charge effects at transition, provided that the magnitude of the jump can be made larger than the γ_{tr} -spread and larger than about 2 to 3-times the (total) change ($2\gamma'$) of γ in the non-adiabatic region.

A jump $\Delta Q_H = -0.25$ will provide almost perfect compensation in the CPS ($\gamma' = 0.055$, $\Delta\gamma_{tr} \approx 0.03$) at present and the booster of the 300 GeV machine ($\gamma' = 0.055$, $\Delta\gamma_{tr} \approx 0.05$).

In the 300 GeV main ring $\gamma' \approx 0.5$ and an unacceptably large Q-jump would be required for compensation of the longitudinal space-charge effect.

Experimental results [4] are in agreement with the predictions of the theory: In the CPS ($\eta_0(0) \approx 1.5$) almost complete compensation was found with a Q-jump of $|\Delta Q_H| \gtrsim 0.2$, $|\dot{\gamma}_{tr}|/\dot{\gamma} \approx 8$. The optimum rf-switching point was found to be close to the middle of the jump region. A "delayed Q-jump" such that the transition point was near to the start of the fast descending branch of the Q-jump waveform, lead to minimum blow up.

TABLE 1: Parameters of a cosine-Q-jump

$$\delta\gamma_{tr} = 0.25 \quad \gamma' = \dot{\gamma}T = 0.05$$

$$\delta x = 0.5 \quad \frac{\langle \dot{\gamma}_{tr} \rangle_{av}}{\dot{\gamma}} = 10$$

Space-charge Parameter $\eta_o(0)$	Optimum delay from start of fast jump transition point $\delta x_{tr}/\delta x$	rf switch $\delta x_{sw}/\delta x$	Bunch area after filamentation A_2/A_1
1.5	0.035	0.3	1.15
3	0.03	0.5	1.2
5	0.02	0.8	2.2

A P P E N D I X 1

Summary of Criteria for the Design of a Q-jump System

We want to summarise some relations relevant to the design of a Q-jump system, which were derived in Section 3 of this report.

In the absence of a detailed theory of the Umstätter effect, we take $\Delta\gamma_{tr} = \Delta Q_{SC}(0)$ as a measure of the spread in γ_{tr} where $\Delta Q_{SC}(0)$ is the single particle Q-shift at transition as obtained from Laslett's formula (more precisely the part that varies with azimuth).

The severity of the space-charge effects discussed in Section 2 and 3 can then be characterized by the two parameters

$$\begin{aligned} \eta_o(0) & \quad : \text{ratio of space-charge defocusing or} \\ & \quad \text{focusing to rf-focusing, calculated from} \\ & \quad \text{bunch length in the absence of space-charge} \\ \Delta x = \frac{\Delta Q_{SC}(0)}{2\gamma'} = \frac{\Delta t}{2T} & : \text{spread in transition energy (-time) compared} \\ & \quad \text{to width of the non-adiabatic region} \end{aligned}$$

The relevance of the parameters T and $\eta_o(0)$ is discussed in refs. [2, 6]. Relations for their calculations are reproduced in Appendix 2 of this report.

Without any compensation, the blow up in area longitudinal space-charge is shown in ref. [6] to be

$$\left(\frac{A_2}{A_1} \right) \approx 1 + 1.4 \eta_o(0)$$

$$(\eta_o \lesssim 1, \text{ for general } \eta \text{ see the graph in [2] for } D = \left(\frac{A_2}{A_1} \right)^{1/2})$$

Similarly the peak momentum deviation (as compared to the low intensity value $(\frac{\Delta P}{P}|0|_{\max})$ is :

$$\left(\frac{\Delta P}{P}\right)_{\max} = \left(\frac{\Delta P}{P}/0\right)_{\max} \cdot 1.18 \eta_0(0)$$

($\eta_0 \gtrsim 1$, for general η see the graph in [8])

The parameter Δx is a measure of the severity of the Umstätter effect. The fractional increase in area may be estimated using

$$\left(\frac{A_2}{A_1}\right) \approx 1 + 1.7 \Delta x \eta_0^{2/3}(0)$$

($\eta_0 \gg 1$, more general relations see (3.24))

To cure the Umstätter effect a Q-jump of (total) width $\delta \gamma_{tr}$ is required with :

$$\delta \gamma_{tr} \gtrsim \Delta Q_{SC}(0)$$

and a speed

$$\left| \frac{\dot{\gamma}_{tr}}{\dot{\gamma}} \right| \gtrsim 1.4 \eta_0(0) \left(\frac{\Delta x}{(A_2/A_1)-1} \right)^{3/2} - 1$$

($\eta_0 \gg 1$ general relation see (3.25))

Here A_2/A_1 is a measure of permissible increase in bunch area.

To compensate longitudinal space-charge, an asymmetric jump is required with

$$\frac{\delta \gamma_{tr}}{\gamma'} \gtrsim 1.4 \eta_o^{4/3}(0)$$

$$\left| \frac{\dot{\gamma}_{tr}}{\dot{\gamma}} \right| \gtrsim 0.68 \left(\frac{(\delta_2 \gamma_{tr})^2 - (\delta_1 \gamma_{tr})^2}{((A_2/A_1) - 1) \gamma'^2} \right)^{3/4}$$

$$((A_2/A_1) - 1) \ll 1$$

Here $\delta_1 \gamma_{tr} + \delta_2 \gamma_{tr} = \delta \gamma_{tr}$ and $\delta_1 \gamma_{tr}$ (Fig. 2) is the distance between the transition point and the start ($\hat{\gamma}_{tr}$) of the jump.

Finally the peak momentum deviation may be estimated using

$$\left(\frac{\Delta P}{P} \right)_{\max} = \left(\frac{\Delta P}{P} / 0 \right)_{\max} \eta_o^{-1/3}(x_1) \left[1 + 0.6 \eta_o(x_1) \frac{\delta \gamma_{tr}}{\gamma'} \left(\frac{\dot{\gamma}}{\dot{\gamma} + (\dot{\gamma}_{tr})} \right)^{3/2} \right]$$

$$\eta_o(x_1) \gg 1$$

$$x_1 < 1$$

where $x_1 = \frac{\delta_1 \gamma_{tr}}{\gamma'}$ is the "normalized time" at the start of the jump. The relation between $\eta_o(x_1)$ and $\eta_o(0)$ can be read from Fig. (5).

The slope ($\dot{\gamma}_{tr}$) of the Q-increase prior to the jump should be somewhat smaller than the slope of the γ -line :

$$\frac{\dot{\gamma}}{\dot{\gamma}_{tr}} \gtrsim 1 + 2 \frac{\gamma'}{\delta \gamma_{tr}} (1 + 0.22 \eta_o(0))$$

The Q-increase after the jump may be parallel to the γ -line.

A P P E N D I X 2

Relations for the Calculation of some Transition Parameters^{*})

$$\eta_o(0) = 19r_p \frac{N g_o}{R A^{3/2}} \left(\frac{f_{rf\infty}}{\dot{\gamma}} \right)$$

r_p : classical particle radius (1.53×10^{-18} m for protons)

N : number of particles per pulse

$g_o \approx 4-5$: geometry factor [6]

R : machine radius

A : bunch area in $\frac{\Delta P}{m_o c}$ x rf-phase units

$f_{rf\infty}$: rf-frequency at $\beta = 1$

$\dot{\gamma}$: $\frac{d\gamma}{dt}$ at transition

$$T = \left(\frac{\gamma^4}{4\pi f_{rf} \dot{\gamma} |\cot \varphi_S| (\dot{\gamma} - \dot{\gamma}_{tr})} \right)^{1/3}$$

all cinemactical quantities (γ, β) taken at transition

f_{rf} : rf-frequency at transition

φ_S : rf-synchronous phase

$\dot{\gamma}_{tr} \approx \dot{Q}_H$: change of γ_{tr} at transition point (due to Q-jump)

$$\frac{\Delta P}{P}/0 = \frac{0.46}{\beta} \left(\frac{A}{\gamma} \right)^{1/2} \left(\frac{\gamma \beta \dot{\gamma} \cot^2 \varphi_S}{\beta f_{rf\infty}} \right)^{1/6}$$

^{*}) This Appendix is based on material prepared by W. Hardt

A P P E N D I X 3

Derivation of an Envelope Equation for the Bunch Length

We want to derive the following envelope equation for the normalized bunch length $\theta(x) \propto \hat{\lambda}(x)$:

$$\frac{d}{dx} \left(\frac{\eta'}{\eta} \frac{d\theta}{dx} \right) + \text{sign}(\eta - \eta_{SW})\theta + \frac{n}{\theta^2} - \frac{\eta}{\eta' \theta^3} = 0 \quad (\text{A3.1})$$

Our derivation follows the derivation of the envelope equation for betatron oscillations given by Courant and Snyder. We attempt to find a solution of the single particle equation (3.2) in the form

$$\lambda \propto \theta(x) e^{i\psi} \quad (\text{A3.2})$$

It is convenient to choose a normalisation such that the area of a matched bunch is π .

Introducing

$$\frac{d\psi}{dx} \equiv \frac{\eta}{\eta' \theta^2} \quad (\text{A3.3})$$

We have then the identity :

$$\frac{d}{dx} \left(\frac{\eta'}{\eta} \frac{d\lambda}{dx} \right) \propto \left[\frac{d}{dx} \left(\frac{\eta'}{\eta} \frac{d\theta}{dx} \right) - \frac{\eta}{\eta' \theta^3} \right] e^{i\psi} \quad (\text{A3.4})$$

Including space-charge by a term $\eta_0(0) \frac{\theta_0^3(0)}{\theta_0^3(x)}$ λ one obtains an equation of the form (A3.1) from (3.2)

Thus by making a "good guess (A3.3)" for the phase ϕ introduced by (A3.2) we arrive at an envelope equation which only contains θ .

To express n by $\eta_0(0)$, note that the phase volume of a matched bunch is (see A3.2) in normalised units

$$\tilde{A} = \pi \theta_{(x)}^2 \quad (\text{A3.5})$$

and in the deviation of (A3.1) it was assumed implicitly that $\tilde{A} = \pi$.

Thus

$$n = \eta_0(0) [\theta_0(0)]^3 = \eta_0(0) \left[\frac{\hat{\lambda}_0(0)}{\sqrt{A/\pi}} \right]^3 \quad (\text{A3.6})$$

where $\hat{\lambda}_0(0)$ is the undisturbed bunch length at transition and A the bunch area. Then following the derivation of ref. [6] one has

$$n = 0.7720 \dots \eta_0(0) \quad (\text{A3.7})$$

Equation (A3.7) may be converted into two first order equation, introducing :

$$P = \frac{\theta'}{\eta/\eta'} \quad (\text{A3.8})$$

One has from (A3.1)

$$\frac{d\theta}{dx} = \frac{\eta}{\eta'} P \quad (\text{A3.9})$$

$$\frac{dP}{dx} = - \text{sign}(\eta - \eta_{SW}) \theta - \frac{n}{\theta^2} + \frac{\eta}{\eta' \theta^3}$$

Going back to the single particle equations (3.2) it can be verified that

$$(\Delta\hat{\beta}\gamma)^2_{\alpha} P^2 = \frac{1}{\theta^2} \quad (\text{A3.10})$$

Thus P and θ are continuous even if η is discontinuous.

For the special case $\frac{\eta}{\eta_1} = x$, the above equations are the same as those derived in ref. [6] (we use n rather than N for the space-charge parameter (A3.6) as we denote by N the number of particles per pulse).

A P P E N D I X 4

Q-Increase prior to the Jump and after the Jump

The speed of these Q-increases is limited by the requirements that the discontinuity in $\hat{\eta}$ does not lead to excessive mismatch. We take the ratio of the equilibrium bunch lengths

$$D = \frac{\hat{\lambda}(x_{Aj})}{\hat{\lambda}(x_A)} \quad (A4.1)$$

as a measure for the mismatch (D^2 is the blow up in area after filamentation). Assuming a linear Q-increase, the constant $\hat{\eta}$ -solution is applicable but one has to take different normalizations for the time in the regions left and right from the discontinuity. By calculations similar to those of Section 3.4 the coordinates of the discontinuity point as measured on these scales can be related to each other.

$$x_{Aj} = x_A \left(\frac{\dot{\gamma}}{\dot{\gamma} - \dot{\gamma}_{tr}} \right)^{2/3}$$

Continuity $\hat{\lambda}(x_A) = \hat{\lambda}(x_{Aj})$ is assured if $x_A \gg 1$ and $x_{Aj} \gg 1$. In fact, in the adiabatic region ($x \gg 1$) from (4.5) :

$$\hat{\lambda}(x) \propto \sqrt[4]{\eta} \quad (A4.2)$$

and no discontinuity occurs if $\eta = \frac{1}{\gamma_{tr}^2} - \frac{1}{\gamma^2}$ is continuous.

Including linear longitudinal space-charge a condition for adiabaticity is $(1 - \eta_{SC})(x_A)^{-1/3} x_A \gg 1$, where

x_A is the normalized time at the start of the slow Q-increase (Fig. 2). Solving for the slope $\dot{\gamma}_{tr}$ of the slow Q-rise, assuming for simplicity $\delta_1 = \delta_2 = \delta \gamma_{tr} / \eta$ (Fig. 2) and admitting $x_{ASC} = 1$:

$$\frac{\dot{\gamma}}{\dot{\gamma}_{tr}} \gtrsim 1 + \frac{2\gamma'}{\delta \gamma_{tr}} (1 - \eta_{SC}(1))^{-1/3} \quad (A4.3)$$

Thus the slope $\dot{\gamma}_{tr}$ should be somewhat smaller than the slope of the γ -curve. The Q-increase after the jump is less critical and can even be parallel to the γ -curve.

A different criterion is obtained assuming a rectangular Q-jump waveform (Fig. 6). Here the mismatch (A4.2) is (using (A4.3)) :

$$D = \left(\frac{\gamma' x_A + \delta \gamma_{tr}}{\gamma' x_A} \right)^{1/4}$$

or

$$x_A \geq \frac{\delta \gamma_{tr} / \gamma'}{D^4 - 1} \quad (A4.4)$$

Taking e.g. $D = 1.1$, $\delta \gamma_{tr} / \gamma' = 4$, $\eta_o(0) = 1.5$ one has $x_A \geq 10$ from (A4.4) and $x_A \approx 1.3$, $\dot{\gamma}_{tr} \approx 0.6\dot{\gamma}$ from (A4.3).

One might think of compensating the mismatch caused by space-charge by the mismatch due to the discontinuities in $\dot{\eta}$ or η . More exotic schemes like "multiple Q-jumping" could be used for the same purpose. But these schemes must all rely on the assumption that there is no important spread in transition time.

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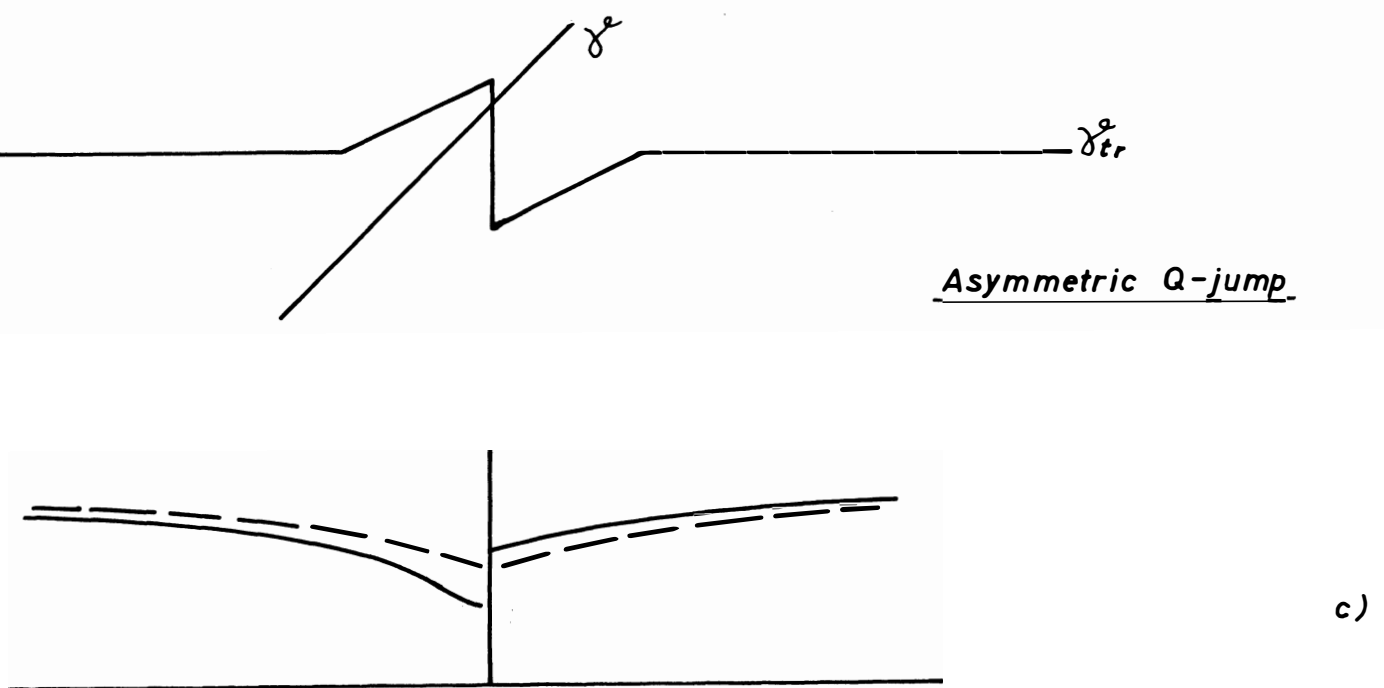
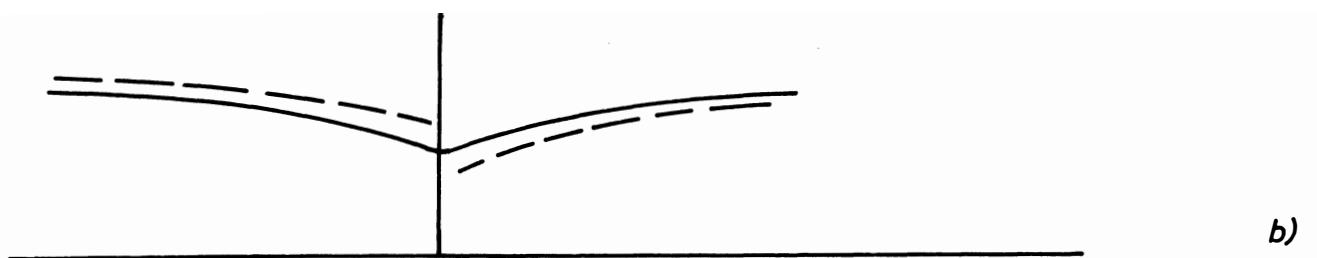
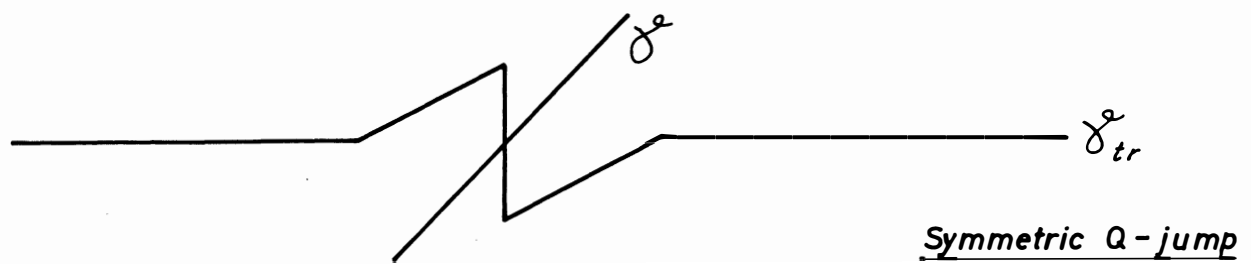
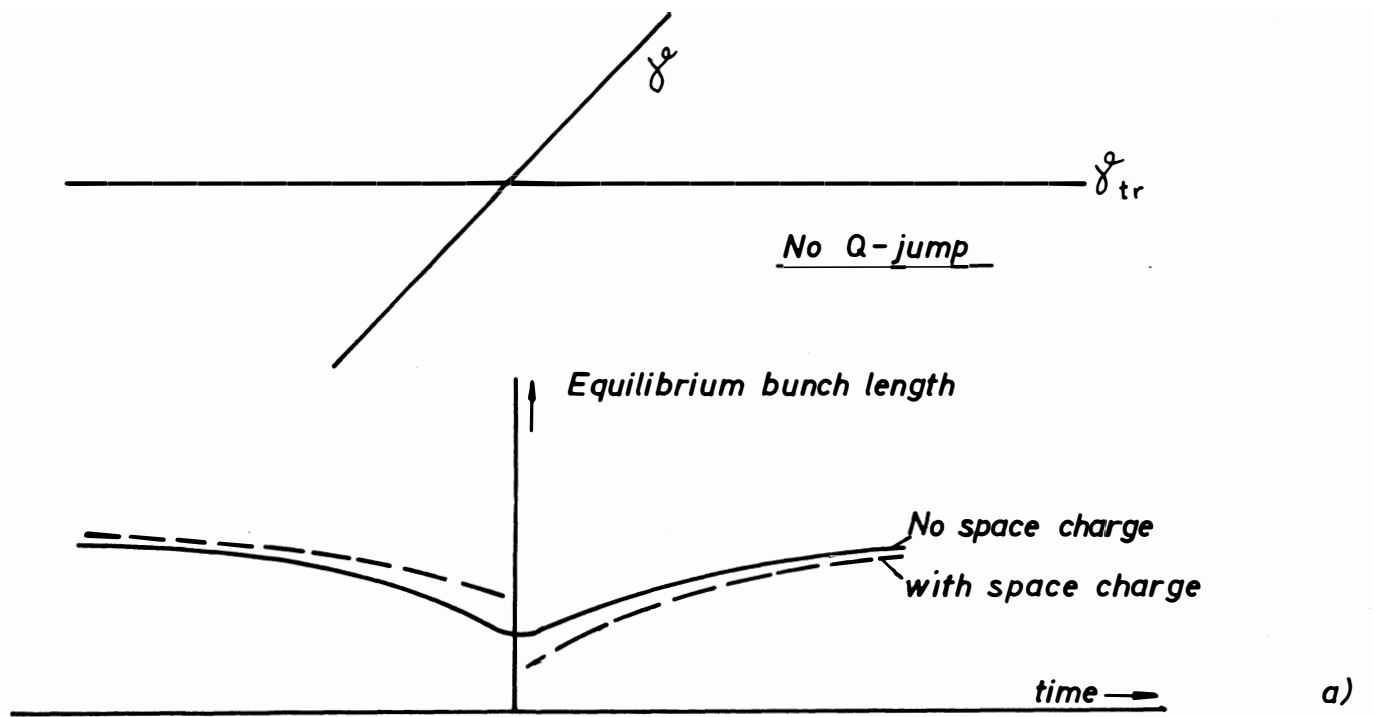


Fig. 1

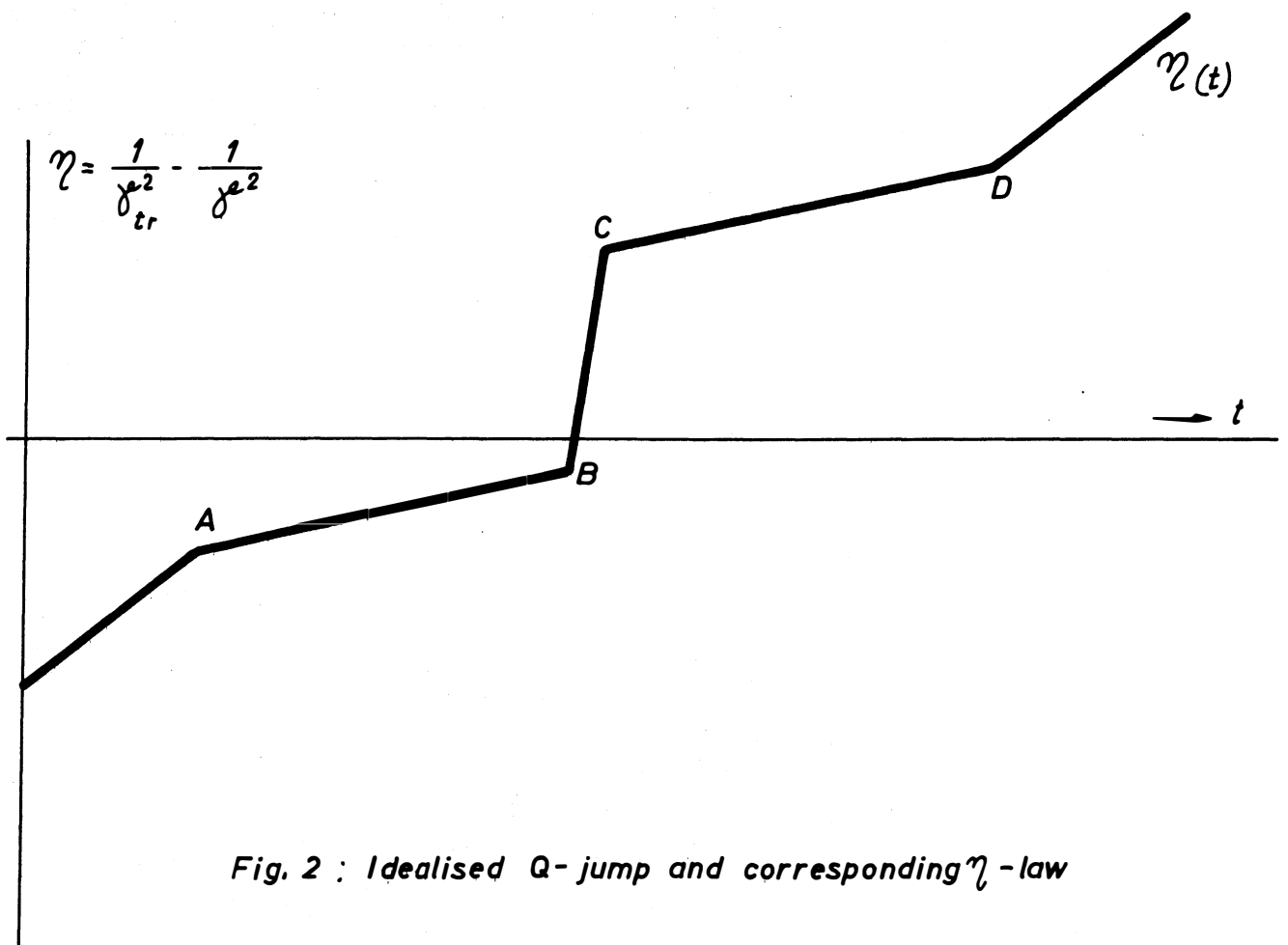
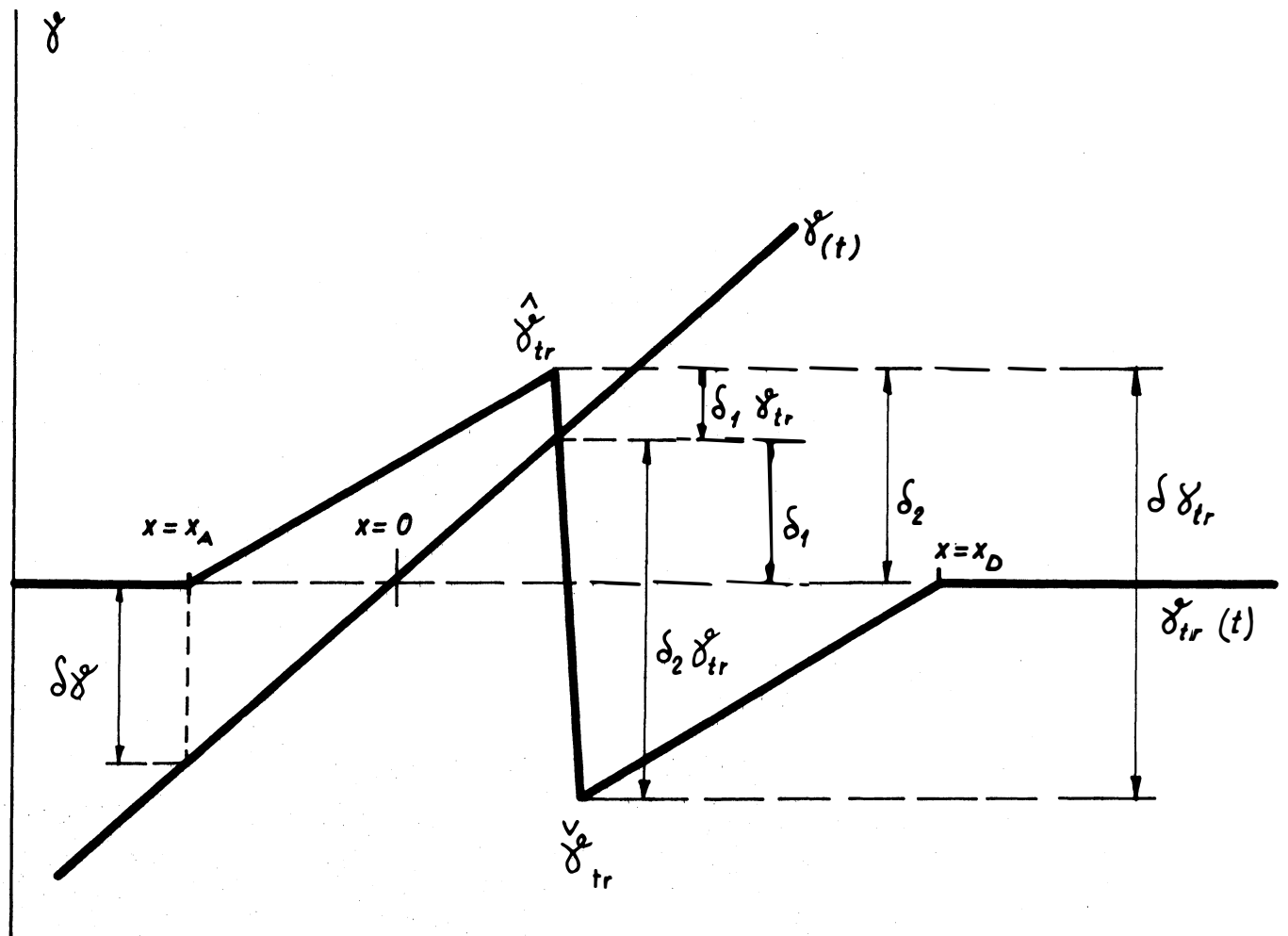


Fig. 2 : Idealised Q-jump and corresponding η -law

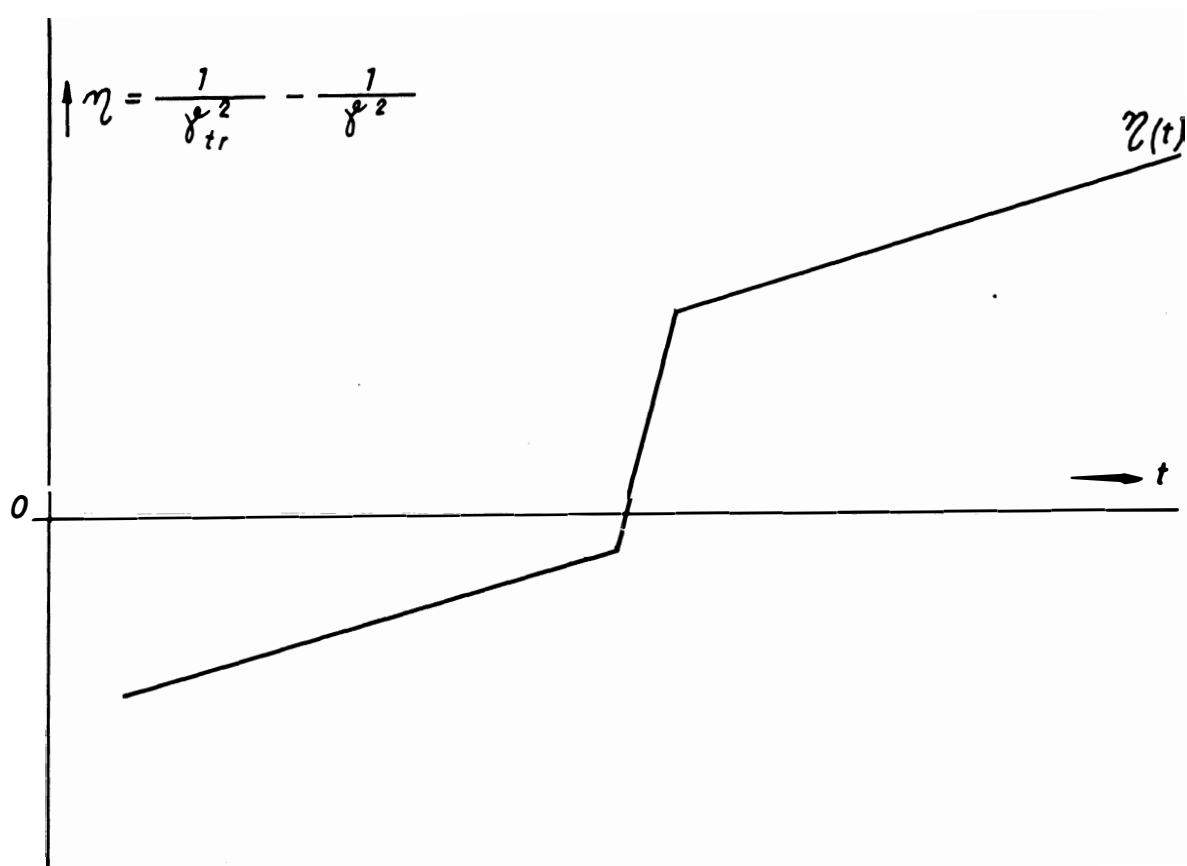


Fig. 3 : Simplified $\eta(t)$ -law

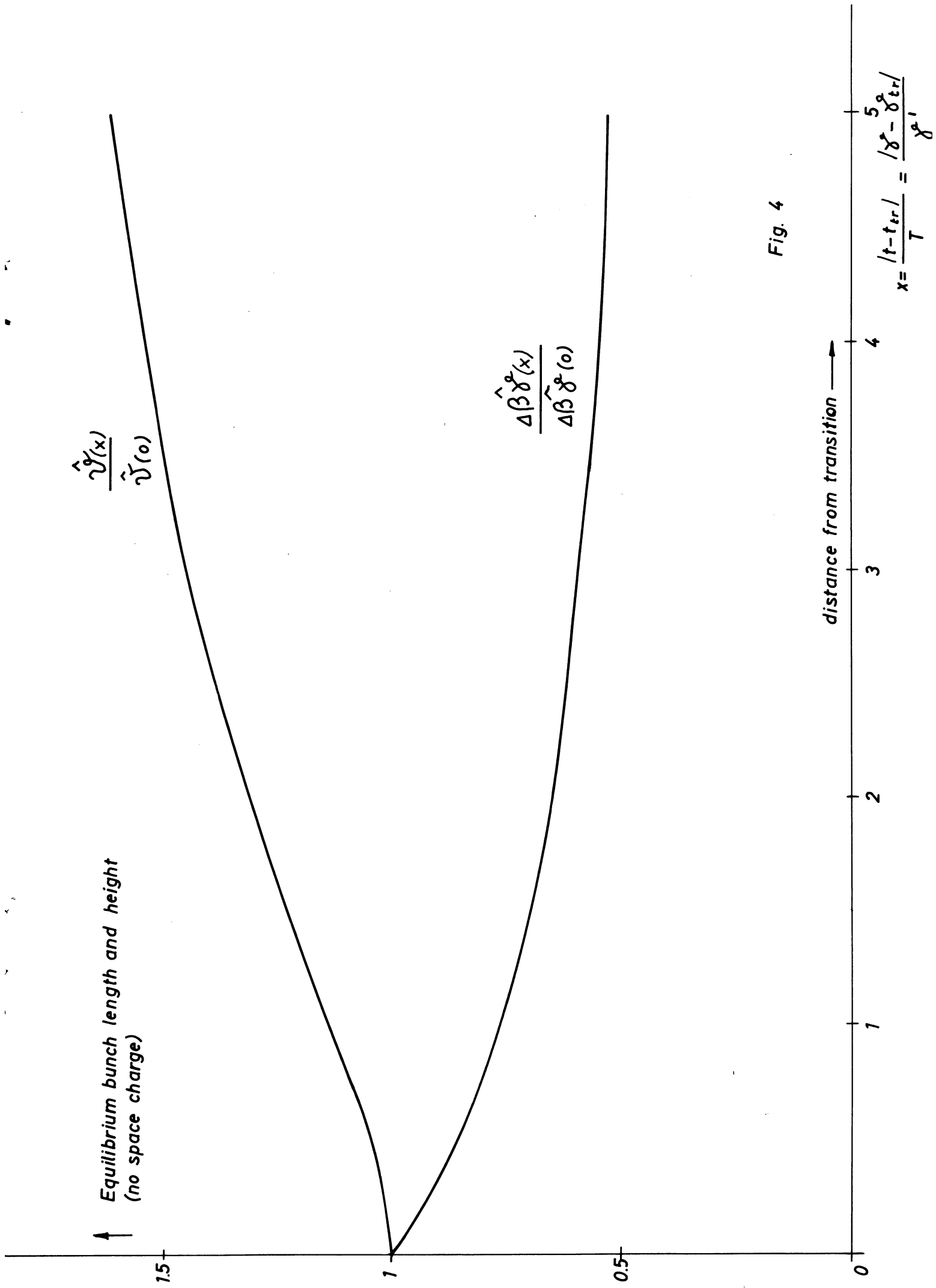


Fig. 4

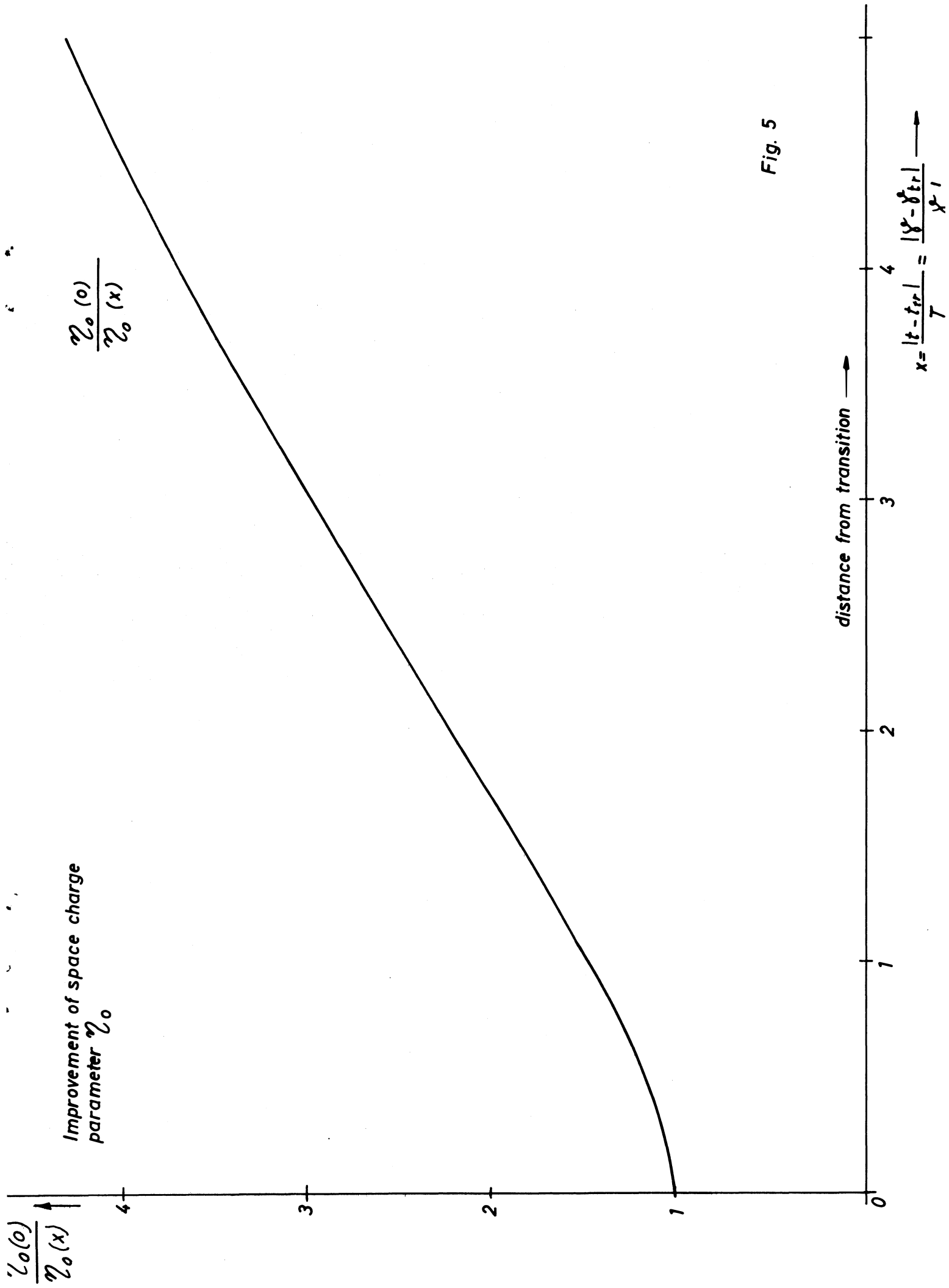


Fig. 5

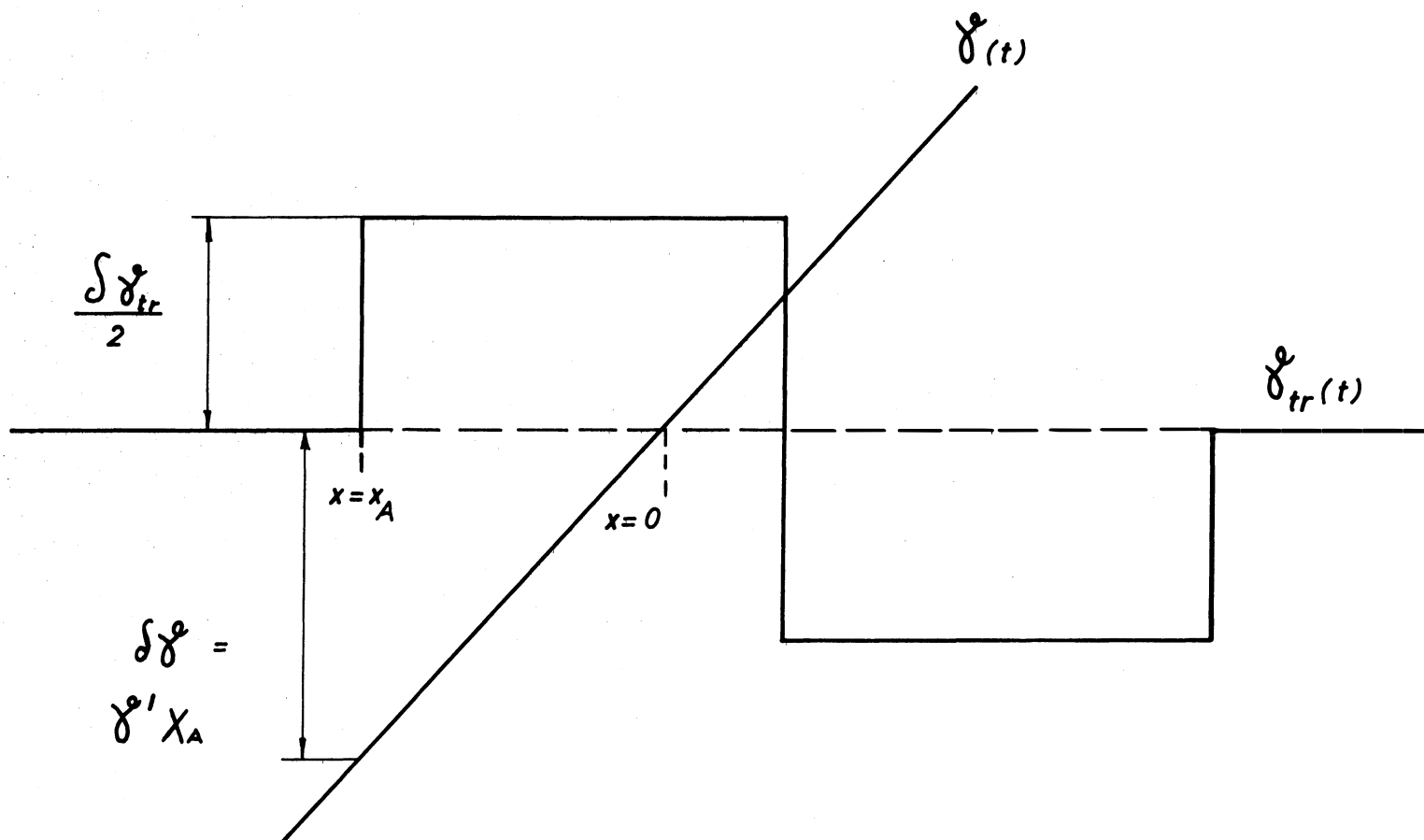


Fig. 6 "Rectangular" Q-jump

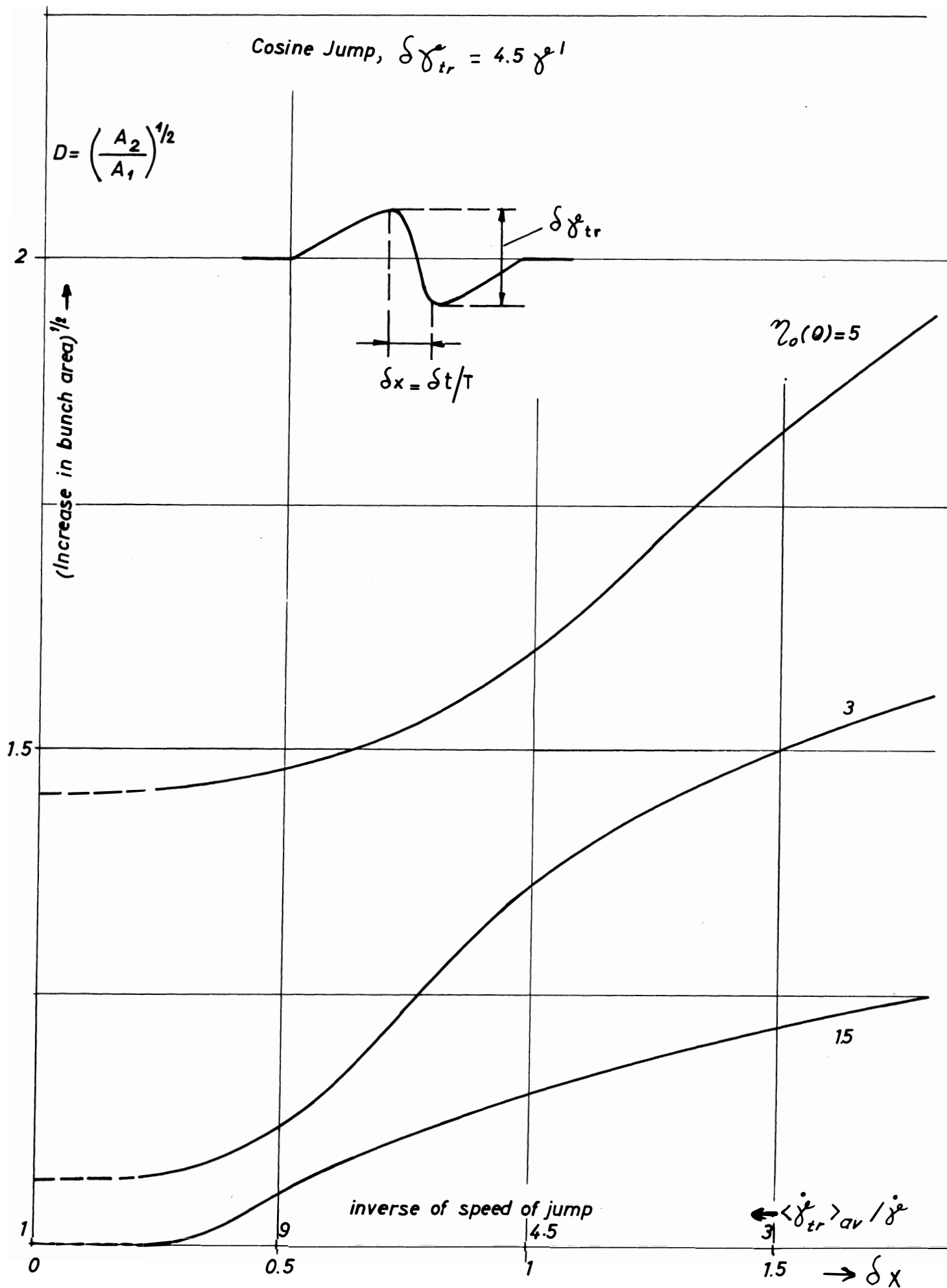


Fig. 7